

## Publication List Leander Thiele

September 16, 2026

- N. Aghanim, B. Maffei, J. Aumont, A. Beelen, B. Borgo, et al. (incl. **L. Thiele**), *FOSSIL: A future mission for CMB spectral distortion measurements*, 2026, [arXiv:2609.xxxxx](#) [[astro-ph.CO](#)]
- B. Cyr, N. Aghanim, E. Baker, E.S. Battistelli, R. Battye, et al. (incl. **L. Thiele**), *Peering Beyond the Veil of Last Scattering: A View of the Universe with CMB Spectral Distortions*, 2026, [arXiv:2609.xxxxx](#) [[astro-ph.CO](#)]
- Y. Itow, J. Liu, H. Maesaka, V. Mikuni, N.-M. Nguyen, et al. (incl. **L. Thiele**), *Future of Artificial Intelligence for Science in Japan 2024 Community Report*, 2026, [arXiv:2608.27807](#) [[hep-ph](#)]
- J. Liu, V. Krishnaraj, K. Vovk, K. Aizawa, A.E. Bayer, et al. (incl. **L. Thiele**), *AI's Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology II: Project Planning and Proposal Evaluation*, 2026, [arXiv:2607.25881](#) [[cs.CL](#)]
- A. Hell, K. Vovk, V. Krishnaraj, J. Liu, K. Aizawa, et al. (incl. **L. Thiele**), *AI's Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology I: Literature Review*, 2026, [arXiv:2607.25672](#) [[astro-ph.IM](#)]
- B.Y. Wang, **L. Thiele**, M. Ho, *Cluster Mass Inference from Galaxy Kinematics*, 2026, [arXiv:2606.12938](#) [[astro-ph.CO](#)]
- L. Thiele**, *Machine Learning Techniques for Astrophysics and Cosmology: Simulation-Based Inference*, 2026, [arXiv:2605.10719](#) [[astro-ph.CO](#)], Invited chapter for the edited book "Machine Learning Techniques for Astrophysics and Cosmology" (editors C. Bambi, V. Kashyap, S. Shashank, N. Yoshida, Springer Singapore, expected in 2026)
- A. Hell, **L. Thiele**, *LLMs with in-context learning for Algorithmic Theoretical Physics*, 2026, [arXiv:2605.08212](#) [[cs.LG](#)], poster at ICML 2026 AI4Physics workshop
- L. Thiele**, *Bayesian Cosmic Void Finding with Graph Flows*, 2026, *OJAp* 9, [arXiv:2602.14630](#) [[astro-ph.CO](#)], poster at ICML 2026 AI4Physics workshop
- J. Armijo, **L. Thiele**, J. Liu, *Replicating weak-lensing summary-statistic covariances with normalizing flows*, 2026, *PRD* 114, 023567, [arXiv:2601.20669](#) [[astro-ph.CO](#)]
- G. Fabbian, F. Bianchini, A. Sabyr, J.C. Hill, C.C. Lovell, **L. Thiele**, D.N. Spergel, *A new constraint on the  $y$ -distortion with FIRAS: implications for feedback models in galaxy formation and cosmic shear measurements*, 2025, [arXiv:2512.03038](#) [[astro-ph.CO](#)]
- A. Tokiwa, A.E. Bayer, J. Armijo, J. Liu, R. Terasawa, **L. Thiele**, M. Alvarez, L. Blot, M. Takada, *Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects*, 2025, *JCAP* 07, 023 [arXiv:2511.20423](#) [[astro-ph.CO](#)],
- B. Barthe-Gold, N.-M. Nguyen, **L. Thiele**, *Reconstructing the local density field with combined convolutional and point cloud architecture*, 2025, [arXiv:2510.08573](#) [[astro-ph.CO](#)], poster at ML4PS workshop NeurIPS 2025

B.Y. Wang, **L. Thiele**, *Set-based Implicit Likelihood Inference of Galaxy Cluster Mass*, 2025, [arXiv:2507.20378](#) [cs.LG], spotlight talk at ML4Astro workshop (colocated with ICML 2025)

J.A. Cowell, J. Armijo, **L. Thiele**, G.A. Marques, C.P. Novaes, D. Grandón, S. Cheng, M. Shirasaki, D. Alonso, J. Liu, *First Constraints from Marked Angular Power Spectra with Subaru Hyper Suprime-Cam Survey First-Year Data*, 2025, MNRAS 546, 2, [arXiv:2507.12315](#) [astro-ph.CO]

**L. Thiele**, A.E. Bayer, N. Takeishi, *Simulation-Efficient Cosmological Inference with Multi-Fidelity SBI*, 2025, [arXiv:2507.00514](#) [astro-ph.CO], poster at ML4Astro workshop (colocated with ICML 2025)

E. Calabrese, J.C. Hill, H.T. Jense, A.L. Posta, I. Abril-Cabezas, et al. (incl. **L. Thiele**), *The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models*, 2025, JCAP 11, 063, [arXiv:2503.14454](#) [astro-ph.CO]

C. Jacobus, **L. Thiele**, P. Harrington, J. Liu, Z. Lukic, *Enhancing Cosmological Simulations with Efficient and Interpretable Machine Learning in the Gabor Wavelet Basis*, 2024, poster at Workshop on Machine Learning and the Physical Sciences (NeurIPS 2024)

**L. Thiele**, *De-baryonifying halos via optimal transport*, 2024, [arXiv:2411.18399](#) [astro-ph.CO]

J. Armijo, G.A. Marques, C.P. Novaes, **L. Thiele**, J.A. Cowell, D. Grandón, M. Shirasaki, J. Liu, *Cosmological constraints using Minkowski functionals from the first year data of the Hyper Suprime-Cam*, 2025, MNRAS 537, 4, [arXiv:2410.00401](#) [astro-ph.CO]

C.P. Novaes, **L. Thiele**, J. Armijo, S. Cheng, J.A. Cowell, G.A. Marques, E.G.M. Ferreira, M. Shirasaki, K. Osato, J. Liu, *Cosmology from HSC Y1 Weak Lensing with Combined Higher-Order Statistics and Simulation-based Inference*, 2025, PRD 111, 8, [arXiv:2409.01301](#) [astro-ph.CO]

A. Kogut, E. Switzer, D. Fixsen, N. Aghanim, J. Chluba, et al. (incl. **L. Thiele**), *The Primordial Inflation Explorer (PIXIE): Mission Design and Science Goals*, 2025, JCAP 004, 020, [arXiv:2405.20403](#) [astro-ph.CO]

S. Cheng, G.A. Marques, D. Grandón, **L. Thiele**, M. Shirasaki, B. Ménard, J. Liu, *Cosmological constraints from weak lensing scattering transform using HSC Y1 data*, 2025, JCAP 01, 006, [arXiv:2404.16085](#) [astro-ph.CO]

D. Grandón, G.A. Marques, **L. Thiele**, S. Cheng, M. Shirasaki, J. Liu, *Impact of baryonic feedback on HSC Y1 weak lensing non-Gaussian statistics*, 2024, PRD 110, 103539, [arXiv:2403.03807](#) [astro-ph.CO]

G.A. Marques, J. Liu, M. Shirasaki, **L. Thiele**, D. Grandón, K.M. Huffenberger, S. Cheng, J. Harnois-Déraps, K. Osato, W.R. Coulton, *Cosmology from weak lensing peaks and minima with Subaru Hyper Suprime-Cam survey first-year data*, 2023, MNRAS 528, 3, [arXiv:2308.10866](#) [astro-ph.CO]

**L. Thiele**, E. Massara, A. Pisani, C. Hahn, D.N. Spergel, S. Ho, B. Wandelt, *Neutrino mass constraint from an Implicit Likelihood Analysis of BOSS voids*, 2024, ApJ 969, 89, [arXiv:2307.07555](#) [astro-ph.CO]

**L. Thiele**, G.A. Marques, J. Liu, M. Shirasaki, *Cosmological constraints from HSC Y1 lensing convergence PDF*, 2023, PRD 108, 123526, [arXiv:2304.05928](#) [astro-ph.CO]

A.M. Delgado, D. Anglés-Alcázar, **L. Thiele**, M. Ntampaka, S. Pandey, K. Lehman, R.S. Somerville, S. Genel, F. Villaescusa-Navarro, *Predicting the impact of feedback on matter clustering with machine learning in CAMELS*, 2023, MNRAS 526, 4, [arXiv:2301.02231](#) [astro-ph.GA]

D. Wadekar, **L. Thiele**, J.C. Hill, S. Pandey, F. Villaescusa-Navarro, et al., *The SZ flux-mass (Y-M) relation at low halo masses: improvements with symbolic regression and strong constraints on baryonic feedback*, 2022, MNRAS 522, 2, [arXiv:2209.02075](#) [astro-ph.CO]

B.K.K. Lee, W. Coulton, **L. Thiele**, S. Ho, *An exploration of the properties of cluster profiles for the thermal and kinetic Sunyaev-Zel'dovich effects*, 2022, MNRAS 517, 420, [arXiv:2205.01710](#) [astro-ph.CO]

**L. Thiele**, M. Cranmer, W. Coulton, S. Ho, D.N. Spergel, *Predicting the Thermal Sunyaev-Zel'dovich Field using Modular and Equivariant Set-Based Neural Networks*, 2022, MLST 3, 035002, [arXiv:2203.00026](#) [astro-ph.CO], poster at the Fourth Workshop on Machine Learning and the Physical Sciences (NeurIPS 2021)

**L. Thiele**, D. Wadekar, J.C. Hill, N. Battaglia, J. Chluba, F. Villaescusa-Navarro, L. Hernquist, M. Vogelsberger, D. Anglés-Alcázar, F. Marinacci, *Percent-level constraints on baryonic feedback with spectral distortion measurements*, 2022, PRD 105, 083505, [arXiv:2201.01663](#) [astro-ph.CO]

D. Wadekar, **L. Thiele**, F. Villaescusa-Navarro, J.C. Hill, D.N. Spergel, M. Cranmer, N. Battaglia, D. Anglés-Alcázar, L. Hernquist, S. Ho, *Augmenting astrophysical scaling relations with machine learning: application to reducing the SZ flux-mass scatter*, 2022, PNAS 120(12), [arXiv:2201.01305](#) [astro-ph.CO]

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, L.A. Perez, P. Villanueva-Domingo, et al. (incl. **L. Thiele**), *The CAMELS project: public data release*, 2022, ApJS 265, 54, [arXiv:2201.01300](#) [astro-ph.CO]

B. Maffei, M.H. Abitbol, N. Aghanim, J. Aumont, E. Battistelli, et al. (incl. **L. Thiele**), *BISO: a balloon project to measure the CMB spectral distortions*, 2021, 16th Marcel Grossmann Meeting, [arXiv:2111.00246](#) [astro-ph.IM]

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, **L. Thiele**, R. Dave, et al., *The CAMELS Multifield Dataset: Learning the Universe's Fundamental Parameters with Artificial Intelligence*, 2021, ApJS 259, 61, [arXiv:2109.10915](#) [cs.LG]

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, D.N. Spergel, Y. Li, et al. (incl. **L. Thiele**), *Robust marginalization of baryonic effects for cosmological inference at the field level*, 2021, [arXiv:2109.10360](#) [astro-ph.CO]

F. Villaescusa-Navarro, D. Anglés-Alcázar, S. Genel, D.N. Spergel, Y. Li, et al. (incl. **L. Thiele**), *Multifield Cosmology with Artificial Intelligence*, 2021, [arXiv:2109.09747](#) [astro-ph.CO]

**L. Thiele**, Y. Guan, J.C. Hill, A. Kosowsky, D.N. Spergel, *Can small-scale baryon inhomogeneities resolve the Hubble tension? An investigation with ACT DR4*, 2021, PRD 104, 063535, [arXiv:2105.03003](#) [astro-ph.CO]

**L. Thiele**, J.C. Hill, K.M. Smith, *Accurate Analytic Model for the Weak Lensing Convergence One-Point Probability Distribution Function and its Auto-Covariance*, 2020, PRD 102, 123545, [arXiv:2009.06547](#) [astro-ph.CO]

**L. Thiele**, F. Villaescusa-Navarro, D.N. Spergel, D. Nelson, A. Pillepich, *Teaching neural networks to generate Fast Sunyaev Zel'dovich Maps*, 2020, ApJ 902, 129, [arXiv:2007.07267 \[astro-ph.CO\]](#)

R. Cayuso, O.J.C. Dias, F. Gray, D. Kubizňák, A. Margalit, J.E. Santos, R.G. Souza, **L. Thiele**, *Massive vector fields in Kerr–Newman and Kerr–Sen black hole spacetimes*, 2020, JHEP 159, [arXiv:1912.08224 \[hep-th\]](#)

**L. Thiele**, C.A.J. Duncan, D. Alonso, *Disentangling magnification in combined shear clustering analyses*, 2020, MNRAS 491, 1746, [arXiv:1907.13205 \[astro-ph.CO\]](#)

R. Cayuso, F. Gray, D. Kubizňák, A. Margalit, R.G. Souza, **L. Thiele**, *Principal Tensor Strikes Again: Separability of Vector Equations with Torsion*, 2019, PLB 795, 650, [arXiv:1906.10072 \[hep-th\]](#)

**L. Thiele**, J.C. Hill, K.M. Smith, *Accurate analytic model for the thermal Sunyaev-Zel'dovich one-point probability distribution function*, 2019, PRD 99, 103511, [arXiv:1812.05584 \[astro-ph.CO\]](#)

F. Dinc, M. Medvidovic, **L. Thiele**, *Effective Geometry Monte Carlo: A Fast and Reliable Simulation Framework for Molecular Communication*, 2019, IEEE Access 7, 28635

F. Dinc, **L. Thiele**, B.C. Akdeniz, *The effective geometry Monte Carlo algorithm: applications to molecular communication*, 2019, PLA 383, 2594, [arXiv:1809.06438 \[cs.ET\]](#)